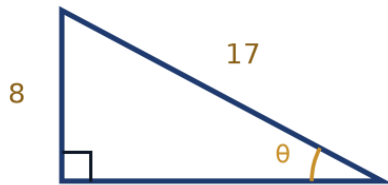


Trigonometry



Right triangle trigonometric ratios

- 1 In a right triangle, angle θ has an opposite side of 8 and hypotenuse 17. Find $\sin \theta$, $\cos \theta$, and $\tan \theta$.

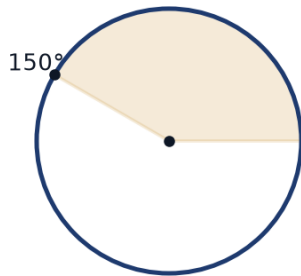


- 2 A ladder 15 feet long leans against a wall, making a 70° angle with the ground. Find the height reached on the wall, to the nearest tenth.

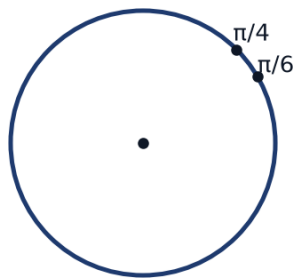


The unit circle and radian measure

- 3 Convert 150° to radians, in terms of π .



- 4 Convert $\frac{3\pi}{4}$ radians to degrees.
- 5 Find the exact value of $\sin\left(\frac{\pi}{6}\right)$ and $\cos\left(\frac{\pi}{4}\right)$, using the unit circle.

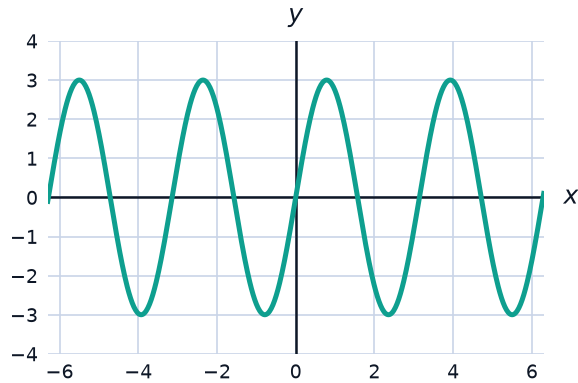


Basic trigonometric identities

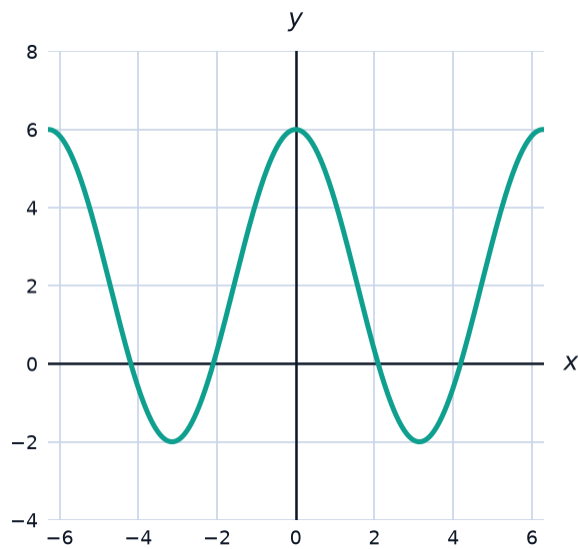
- 6 Use the Pythagorean identity to find $\cos \theta$, given that $\sin \theta = \frac{3}{5}$ and θ is in the first quadrant.
- 7 Simplify $\frac{\sin \theta}{\cos \theta} \cdot \cos \theta$.

Graphs of sine and cosine functions

- 8 State the amplitude and period of $y = 3\sin(2x)$.

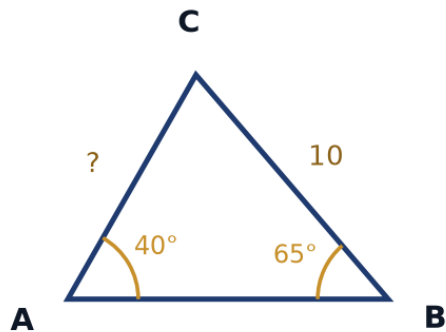


- 9 State the amplitude, period, and vertical shift of $y = 4\cos(x) + 2$.



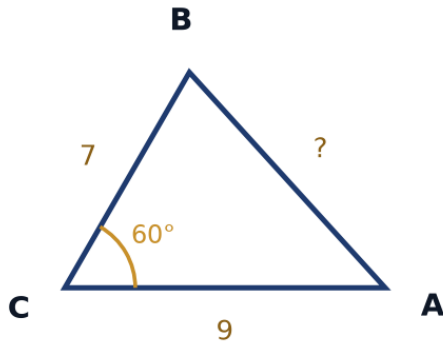
Law of Sines

- 10 In triangle ABC , $A = 40^\circ$, $B = 65^\circ$, and $a = 10$. Find side b , to the nearest tenth, using the Law of Sines.



Law of Cosines

- 11 In triangle ABC , $a = 7$, $b = 9$, and $C = 60^\circ$. Find side c , using the Law of Cosines.

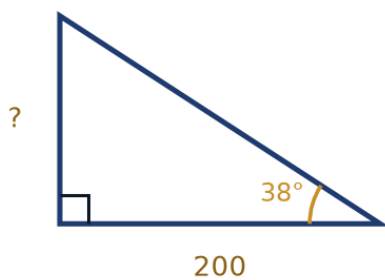


Trigonometric equations

- 12 Solve $2\sin \theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

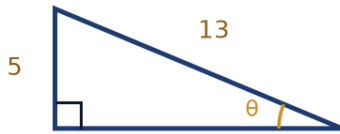
Applications of trigonometry to real-world problems

- 13 This question has two parts. A surveyor stands 200 feet from the base of a tower and measures the angle of elevation to the top as 38° . (a) Find the height of the tower, to the nearest foot. (b) If the surveyor moves 50 feet closer to the tower, find the new angle of elevation, to the nearest degree, using the height found in part (a).

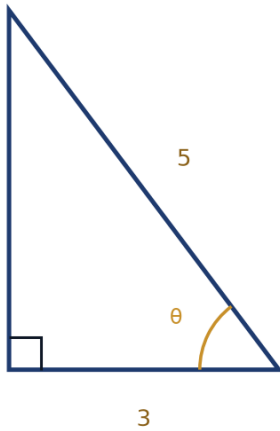


Reciprocal trigonometric functions

- 14 If $\sin \theta = \frac{5}{13}$, find $\csc \theta$.

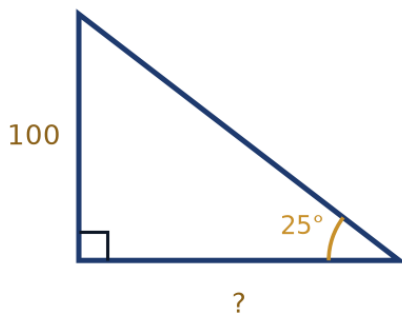


- 15 If $\cos \theta = \frac{3}{5}$, find $\sec \theta$ and $\tan \theta$ (assuming θ is in the first quadrant).



Angle of elevation and depression

- 16 From the top of a 100-foot cliff, the angle of depression to a boat is 25° . Find the horizontal distance from the base of the cliff to the boat, to the nearest foot.

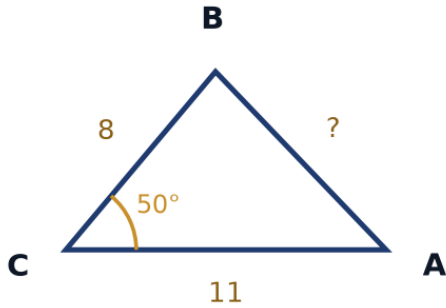


Solving trigonometric equations over a restricted domain

- 17 Solve $2\cos \theta + 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

A multi-part triangle problem combining Law of Sines and Law of Cosines

- 18** This question has two parts. Triangle ABC has $a = 8$, $b = 11$, and $C = 50^\circ$. (a) Use the Law of Cosines to find side c , to the nearest tenth. (b) Use the Law of Sines to find angle A , to the nearest degree.



Cofunction and negative angle identities

- 19** Use the cofunction identity to find $\cos(30^\circ)$ given that $\sin(60^\circ) = \frac{\sqrt{3}}{2}$.
- 20** Use the fact that sine is an odd function to find $\sin(-30^\circ)$, given $\sin(30^\circ) = \frac{1}{2}$.

Multiple choice — easy

- 21** In a right triangle, the angle θ has an opposite side of length 20, adjacent side 21, and hypotenuse 29. Find $\tan \theta$.
- a. $\frac{29}{21}$
 - b. $\frac{20}{21}$
 - c. $\frac{29}{20}$
 - d. $\frac{21}{20}$

- 22 In a right triangle, the angle θ has an opposite side of length 9, adjacent side 12, and hypotenuse 15. Find $\sin \theta$.
- a. $\frac{4}{3}$
 - b. $\frac{3}{5}$
 - c. $\frac{5}{4}$
 - d. $\frac{5}{3}$
- 23 In a right triangle, the angle θ has an opposite side of length 8, adjacent side 15, and hypotenuse 17. Find $\tan \theta$.
- a. $\frac{17}{15}$
 - b. $\frac{15}{8}$
 - c. $\frac{17}{8}$
 - d. $\frac{8}{15}$
- 24 In a right triangle, the angle θ has an opposite side of length 20, adjacent side 21, and hypotenuse 29. Find $\cos \theta$.
- a. $\frac{29}{20}$
 - b. $\frac{21}{20}$
 - c. $\frac{21}{29}$
 - d. $\frac{29}{21}$
- 25 In a right triangle, the angle θ has an opposite side of length 3, adjacent side 4, and hypotenuse 5. Find $\tan \theta$.
- a. $\frac{5}{4}$
 - b. $\frac{5}{3}$
 - c. $\frac{3}{4}$
 - d. $\frac{4}{3}$

- 26 In a right triangle, the angle θ has an opposite side of length 9, adjacent side 12, and hypotenuse 15. Find $\cos \theta$.
- a. $\frac{5}{4}$
 - b. $\frac{5}{3}$
 - c. $\frac{4}{5}$
 - d. $\frac{4}{3}$
- 27 In a right triangle, the angle θ has an opposite side of length 5, adjacent side 12, and hypotenuse 13. Find $\sin \theta$.
- a. $\frac{13}{12}$
 - b. $\frac{13}{5}$
 - c. $\frac{5}{13}$
 - d. $\frac{12}{5}$
- 28 In a right triangle, the angle θ has an opposite side of length 12, adjacent side 16, and hypotenuse 20. Find $\tan \theta$.
- a. $\frac{4}{3}$
 - b. $\frac{3}{4}$
 - c. $\frac{5}{3}$
 - d. $\frac{5}{4}$
- 29 In a right triangle, the angle θ has an opposite side of length 9, adjacent side 40, and hypotenuse 41. Find $\sin \theta$.
- a. $\frac{40}{9}$
 - b. $\frac{41}{40}$
 - c. $\frac{9}{41}$
 - d. $\frac{41}{9}$

- 30 In a right triangle, the angle θ has an opposite side of length 9, adjacent side 12, and hypotenuse 15. Find $\tan \theta$.
- a. $\frac{5}{4}$
 - b. $\frac{5}{3}$
 - c. $\frac{3}{4}$
 - d. $\frac{4}{3}$
- 31 In a right triangle, the angle θ has an opposite side of length 5, adjacent side 12, and hypotenuse 13. Find $\cos \theta$.
- a. $\frac{13}{12}$
 - b. $\frac{12}{5}$
 - c. $\frac{13}{5}$
 - d. $\frac{12}{13}$
- 32 In a right triangle, the angle θ has an opposite side of length 12, adjacent side 16, and hypotenuse 20. Find $\cos \theta$.
- a. $\frac{5}{4}$
 - b. $\frac{5}{3}$
 - c. $\frac{4}{5}$
 - d. $\frac{4}{3}$
- 33 In a right triangle, the angle θ has an opposite side of length 8, adjacent side 15, and hypotenuse 17. Find $\sin \theta$.
- a. $\frac{17}{15}$
 - b. $\frac{17}{8}$
 - c. $\frac{8}{17}$
 - d. $\frac{15}{8}$

- 34 In a right triangle, the angle θ has an opposite side of length 7, adjacent side 24, and hypotenuse 25. Find $\sin \theta$.
- a. $\frac{7}{25}$
 - b. $\frac{25}{7}$
 - c. $\frac{24}{7}$
 - d. $\frac{25}{24}$
- 35 In a right triangle, the angle θ has an opposite side of length 12, adjacent side 16, and hypotenuse 20. Find $\sin \theta$.
- a. $\frac{5}{4}$
 - b. $\frac{4}{3}$
 - c. $\frac{3}{5}$
 - d. $\frac{5}{3}$
-

Multiple choice — medium

- 36 Convert 170° to radians, in terms of π .
- a. $\frac{18}{17}\pi$
 - b. $\frac{17}{18}\pi$
 - c. $\frac{10}{9}\pi$
 - d. $\frac{17}{9}\pi$
- 37 Convert 140° to radians, in terms of π .
- a. $\frac{14}{9}\pi$
 - b. $\frac{17}{18}\pi$
 - c. $\frac{9}{7}\pi$
 - d. $\frac{7}{9}\pi$

38 Convert 350° to radians, in terms of π .

a. $\frac{35}{18}\pi$

b. $\frac{18}{35}\pi$

c. $\frac{19}{9}\pi$

d. $\frac{35}{9}\pi$

39 Convert 108° to radians, in terms of π .

a. $\frac{6}{5}\pi$

b. $\frac{3}{5}\pi$

c. $\frac{5}{3}\pi$

d. $\frac{23}{30}\pi$

40 Convert 60° to radians, in terms of π .

a. $\frac{2}{3}\pi$

b. $\frac{1}{3}\pi$

c. $\frac{1}{2}\pi$

d. 3π

41 Convert 144° to radians, in terms of π .

a. $\frac{5}{4}\pi$

b. $\frac{4}{5}\pi$

c. $\frac{29}{30}\pi$

d. $\frac{8}{5}\pi$

42 Convert 210° to radians, in terms of π .

a. $\frac{4}{3}\pi$

b. $\frac{6}{7}\pi$

c. $\frac{7}{3}\pi$

d. $\frac{7}{6}\pi$

43 Convert 240° to radians, in terms of π .

a. $\frac{4}{3}\pi$

b. $\frac{3}{4}\pi$

c. $\frac{3}{2}\pi$

d. $\frac{8}{3}\pi$

44 Convert 45° to radians, in terms of π .

a. $\frac{5}{12}\pi$

b. $\frac{1}{2}\pi$

c. 4π

d. $\frac{1}{4}\pi$

45 Convert 90° to radians, in terms of π .

a. π

b. $\frac{1}{2}\pi$

c. $\frac{2}{3}\pi$

d. 2π

46 Convert 330° to radians, in terms of π .

a. $\frac{11}{3}\pi$

b. $\frac{11}{6}\pi$

c. $\frac{6}{11}\pi$

d. 2π

47 Convert 15° to radians, in terms of π .

a. 12π

b. $\frac{1}{12}\pi$

c. $\frac{1}{4}\pi$

d. $\frac{1}{6}\pi$

48 Convert 300° to radians, in terms of π .

a. $\frac{5}{3}\pi$

b. $\frac{10}{3}\pi$

c. $\frac{3}{5}\pi$

d. $\frac{11}{6}\pi$

49 Convert 160° to radians, in terms of π .

a. $\frac{9}{8}\pi$

b. $\frac{16}{9}\pi$

c. $\frac{19}{18}\pi$

d. $\frac{8}{9}\pi$

50 Convert 180° to radians, in terms of π .

a. $\frac{13}{12}\pi$

b. $\frac{7}{6}\pi$

c. 2π

d. π

Multiple choice — hard

51 In triangle ABC , $A = 32^\circ$, $B = 57^\circ$, and $a = 9$. Find side b , to the nearest tenth, using the Law of Sines.

a. 14.2

b. 12.2

c. 16.2

d. 18.5

52 In triangle ABC , $A = 61^\circ$, $B = 29^\circ$, and $a = 6$. Find side b , to the nearest tenth, using the Law of Sines.

a. 4.3

b. 1.3

c. 3.3

d. 5.3

53 In triangle ABC , $A = 48^\circ$, $B = 34^\circ$, and $a = 5$. Find side b , to the nearest tenth, using the Law of Sines.

a. 3.8

b. 1.8

c. 5.8

d. 4.9

- 54 In triangle ABC , $A = 43^\circ$, $B = 47^\circ$, and $a = 14$. Find side b , to the nearest tenth, using the Law of Sines.
- a. 19.5
 - b. 17.0
 - c. 15.0
 - d. 13.0
- 55 In triangle ABC , $A = 34^\circ$, $B = 36^\circ$, and $a = 17$. Find side b , to the nearest tenth, using the Law of Sines.
- a. 17.9
 - b. 23.2
 - c. 15.9
 - d. 19.9
- 56 In triangle ABC , $a = 14$, $b = 7$, and $C = 105^\circ$. Find side c , to the nearest tenth, using the Law of Cosines.
- a. 18.7
 - b. 17.2
 - c. 20.6
 - d. 15.7
- 57 In triangle ABC , $a = 10$, $b = 12$, and $C = 105^\circ$. Find side c , to the nearest tenth, using the Law of Cosines.
- a. 17.5
 - b. 21.0
 - c. 16.0
 - d. 19.0

- 58 In triangle ABC , $a = 15$, $b = 11$, and $C = 120^\circ$. Find side c , to the nearest tenth, using the Law of Cosines.
- a. 27.1
 - b. 22.6
 - c. 21.1
 - d. 24.1
- 59 In triangle ABC , $a = 6$, $b = 12$, and $C = 30^\circ$. Find side c , to the nearest tenth, using the Law of Cosines.
- a. 5.9
 - b. 7.4
 - c. 7.9
 - d. 8.9
- 60 In triangle ABC , $a = 8$, $b = 15$, and $C = 135^\circ$. Find side c , to the nearest tenth, using the Law of Cosines.
- a. 19.9
 - b. 25.7
 - c. 21.4
 - d. 22.9
- 61 Solve $2\cos\theta + 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.
- a. $45^\circ, 60^\circ$
 - b. $60^\circ, 240^\circ$
 - c. $180^\circ, 225^\circ$
 - d. $120^\circ, 240^\circ$

62 Solve $2\cos\theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

a. $270^\circ, 300^\circ$

b. $30^\circ, 210^\circ$

c. $45^\circ, 315^\circ$

d. $60^\circ, 300^\circ$

63 Solve $2\sin\theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

a. $45^\circ, 150^\circ$

b. $30^\circ, 150^\circ$

c. $120^\circ, 315^\circ$

d. $30^\circ, 300^\circ$

64 Solve $\tan\theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

a. $240^\circ, 300^\circ$

b. $45^\circ, 225^\circ$

c. $30^\circ, 225^\circ$

d. $135^\circ, 270^\circ$

65 Solve $\tan\theta + 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

a. $135^\circ, 315^\circ$

b. $135^\circ, 240^\circ$

c. $60^\circ, 240^\circ$

d. $90^\circ, 300^\circ$